

Learning objectives

In this lesson you will learn to:

- apply arithmetic left and right shifts to signed binary numbers
- describe how an arithmetic right shift differs from a logical right shift.

For more information about this topic and additional student activities see Topic 2 of the student book.

Left arithmetic shifting

Shifts are known as 'bitwise operations' because they operate on each bit in a pattern, one at a time.

Left-shift 0 0 0 1 0 1 1 1 (decimal 23)



= 0 0 1 0 1 1 1 0 ← 0 (decimal 46)

The Most Significant Bit (MSB) is shifted left, out of the pattern. Each remaining bit is shifted left to the next location.

A new 0 is placed into the vacated location on the right (the Least Significant Bit - LSB).

This has the effect of multiplying by 2.

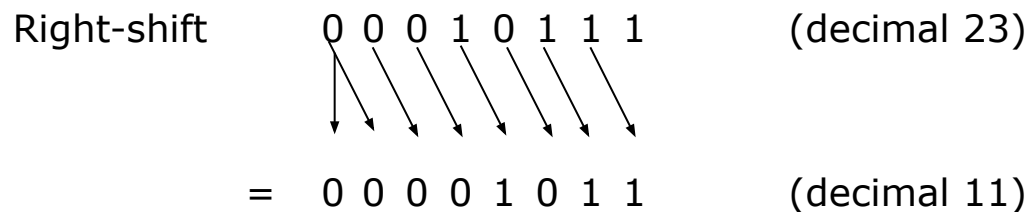
Left arithmetic shifting is the same as left logical shifting.

Right arithmetic shifting

The LSB is shifted right, out of the pattern.

The most significant bit (MSB) value is **always** maintained. It is copied into the vacated location on the left. In this example a 0 is inserted.

This has the effect of dividing by 2, while maintaining the sign.



In multiple shifts, the MSB is **duplicated** and inserted into **all** empty spaces.

Right arithmetic shifting is different to right logical shifting as, instead of a 0 being inserted, the value of the most significant bit is maintained.

For this example, and for most unsigned integers, the same effect is produced. You will see a difference with signed integers.

Signed integers

Let's look at a signed integer and perform a left arithmetic shift of 3.

The signed integer is a two's complement binary of -22:

1	1	1	0	1	0	1	0
---	---	---	---	---	---	---	---

First shift

1	1	0	1	0	1	0	0
---	---	---	---	---	---	---	---

-44

Second shift

1	0	1	0	1	0	0	0
---	---	---	---	---	---	---	---

-88

Third shift

0	1	0	1	0	0	0	0
---	---	---	---	---	---	---	---

80

Is there a problem here?
-88 x 2 should be -176

Overflow

We've looked at overflows before.

An overflow error is when the result of a calculation is too large for the space allocated for it.

We've just seen the effect that overflows have when performing shifts.

Wrap up: you have learned how to...

- ✓ Apply arithmetic left shifts to signed binary numbers:
 - Arithmetic left shifting is the same as logical left shifting.
 - Overflow is likely when shifting signed integers.
- ✓ Apply arithmetic right shifts to signed binary numbers.
- ✓ Explain how an arithmetic shift right differs from a logical shift right.
 - Arithmetic right shifting maintains the MSB and duplicates it in every vacant position.
 - Logical right shifting fills in with 0s on the left.